

Question:

To design and implement an UPI payment system in java using interfaces ,classes and objects,demonstrating encapsulation,abstraction through interfaces,constructors,method overriding,custom exception handling,validation,wallet balance management and payment processing.

System Requirements:

Hardware Requirements:

1. Standard PC/Laptop
2. Intel Core i3/i5 processor or equivalent
3. Minimum 4 GB RAM
4. Minimum 500 MB free storage

Software Requirements:

1. Operating System: Windows / Linux / macOS
2. Java Development Kit (JDK 17 or JDK 21)
3. IDE such as Eclipse, IntelliJ IDEA, NetBeans, or VS Code
4. Command Prompt / Terminal

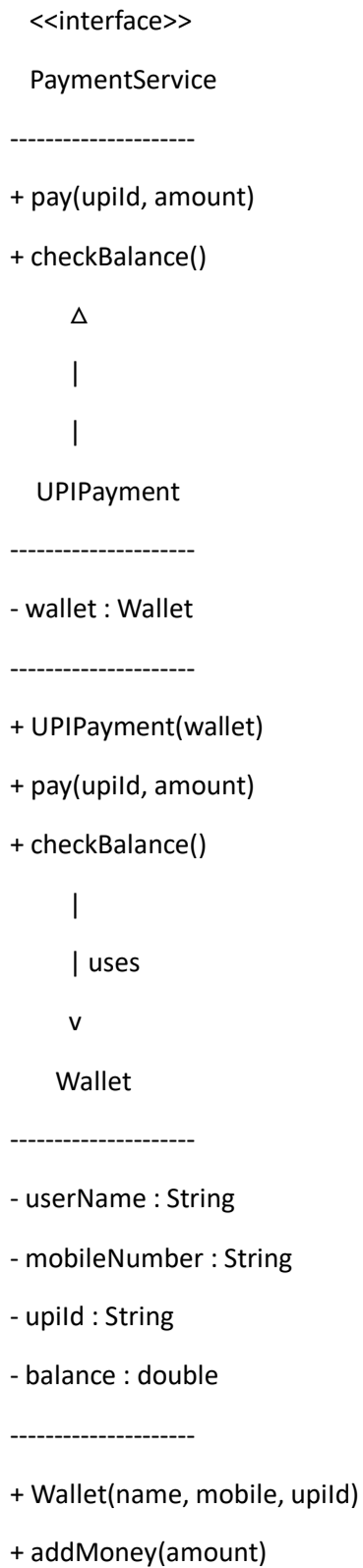
Execution Steps:

1. Start the program.
2. Create a Wallet object for Lalitha with an initial balance of ₹0.0.
3. Display the wallet details.
4. Add ₹5000 to the wallet.
5. Create a UPIPayment object using the wallet.
6. Display the available balance: ₹5000.0.
7. Make a payment of ₹1500 to rahul@upi.
8. Validate the UPI ID, payment amount, and wallet balance.
9. Deduct ₹1500 and display the successful payment message.
10. Display the remaining balance: ₹3500.0.
11. Try paying with an invalid UPI ID; display the error message.
12. Try paying a negative amount; display the error message.
13. Try paying ₹5000 with only ₹3500 available; display the insufficient balance message.

14. Display the final wallet details with a balance of ₹3500.0.

15. Stop the program.

UML Class Diagram:



```
+ getBalance() : double
+ deductMoney(amount)
+ displayWalletDetails()
```

```
<<exception>>      <<exception>>
InsufficientBalanceException  InvalidUPIException
-----
+ constructor(message)  + constructor(message)
```

```
<<exception>>
InvalidAmountException
-----
+ constructor(message)
```

Code:

```
// PaymentService Interface
interface PaymentService {

    void pay(String upid, double amount);

    void checkBalance();
}

// Custom Exception for insufficient balance
class InsufficientBalanceException extends Exception {

    public InsufficientBalanceException(String message) {
        super(message);
    }
}
```

```
}  
}
```

```
// Custom Exception for invalid UPI ID
```

```
class InvalidUPIException extends Exception {
```

```
    public InvalidUPIException(String message) {  
        super(message);  
    }  
}
```

```
// Custom Exception for invalid amount
```

```
class InvalidAmountException extends Exception {
```

```
    public InvalidAmountException(String message) {  
        super(message);  
    }  
}
```

```
// Wallet Class
```

```
class Wallet {
```

```
    // Private data members to demonstrate encapsulation
```

```
    private String userName;  
    private String mobileNumber;  
    private String upid;  
    private double balance;
```

```
// Constructor  
public Wallet(String userName, String mobileNumber, String upild) {  
    this.userName = userName;  
    this.mobileNumber = mobileNumber;  
    this.upild = upild;  
    this.balance = 0.0;  
}
```

```
// Method to add money  
public void addMoney(double amount) {  
  
    if (amount > 0) {  
        balance = balance + amount;  
        System.out.println("Money added successfully: ₹" + amount);  
    } else {  
        System.out.println("Invalid amount.");  
    }  
}
```

```
// Method to get balance  
public double getBalance() {  
    return balance;  
}
```

```
// Method to deduct money  
public void deductMoney(double amount) {  
    balance = balance - amount;  
}
```

```
// Method to display wallet details  
public void displayWalletDetails() {
```

```
        System.out.println("\n----- Wallet Details -----");
        System.out.println("User Name   : " + userName);
        System.out.println("Mobile Number: " + mobileNumber);
        System.out.println("UPI ID     : " + upid);
        System.out.println("Balance    : ₹" + balance);
    }
}
```

// UPI Payment Class

```
class UPIPayment implements PaymentService {
```

```
    private Wallet wallet;
```

// Constructor

```
    public UPIPayment(Wallet wallet) {
        this.wallet = wallet;
    }
```

// Pay method

@Override

```
    public void pay(String upid, double amount) {
```

```
        try {
```

```
            // Validate UPI ID
```

```
            if (upid == null ||
```

```
                !upid.contains("@") ||
```

```
                upid.startsWith("@") ||
```

```
                upid.endsWith("@")) {
```

```
        throw new InvalidUPIException(
            "Invalid UPI ID: " + upild
        );
    }

    // Validate payment amount
    if (amount <= 0) {

        throw new InvalidAmountException(
            "Payment amount must be greater than zero."
        );
    }

    // Check sufficient balance
    if (amount > wallet.getBalance()) {

        throw new InsufficientBalanceException(
            "Insufficient balance for this payment."
        );
    }

    // Deduct payment amount
    wallet.deductMoney(amount);

    System.out.println("\nPayment Successful!");
    System.out.println("Receiver UPI ID: " + upild);
    System.out.println("Amount Paid    : ₹" + amount);

    }
```

```

catch (InvalidUPIException e) {

    System.out.println("Transaction Failed: "
        + e.getMessage());

}

catch (InvalidAmountException e) {

    System.out.println("Transaction Failed: "
        + e.getMessage());

}

catch (InsufficientBalanceException e) {

    System.out.println("Transaction Failed: "
        + e.getMessage());

}

finally {

    System.out.println("Transaction processing completed.");
}

}

// Check balance method
@Override
public void checkBalance() {

```

```
        System.out.println(
            "Available Wallet Balance: ₹"
            + wallet.getBalance()
        );
    }
}
```

```
// Main Class
```

```
public class UpiPayment {
```

```
    public static void main(String[] args) {
```

```
        // Create Wallet object
```

```
        Wallet wallet = new Wallet(
```

```
            "Lalitha",
```

```
            "987673210",
```

```
            "lalitha@upi"
```

```
        );
```

```
        // Display initial wallet details
```

```
        wallet.displayWalletDetails();
```

```
        // Add money to wallet
```

```
        wallet.addMoney(5000);
```

```
        // Create UPI Payment object
```

```
        PaymentService payment = new UPIPayment(wallet);
```

```
        // Check balance
```

```
payment.checkBalance();

// Make a valid payment
System.out.println("\n--- Valid Payment ---");
payment.pay("rahul@upi", 1500);

// Check balance after payment
payment.checkBalance();

// Invalid UPI test
System.out.println("\n--- Invalid UPI Test ---");
payment.pay("rahulupi", 500);

// Invalid amount test
System.out.println("\n--- Invalid Amount Test ---");
payment.pay("rahul@upi", -100);

// Insufficient balance test
System.out.println("\n--- Insufficient Balance Test ---");
payment.pay("rahul@upi", 5000);

// Display final wallet details
wallet.displayWalletDetails();
}
}
```

Output:

----- Wallet Details -----

User Name : Lalitha

Mobile Number: 9876732100

UPI ID : lalitha@upi

Balance : ₹0.0

Money added successfully: ₹5000.0

Available Wallet Balance: ₹5000.0

--- Valid Payment ---

Payment Successful!

Receiver UPI ID: rahul@upi

Amount Paid : ₹1500.0

Transaction processing completed.

Available Wallet Balance: ₹3500.0

--- Invalid UPI Test ---

Transaction Failed: Invalid UPI ID: rahulupi

Transaction processing completed.

--- Invalid Amount Test ---

Transaction Failed: Payment amount must be greater than zero.

Transaction processing completed.

--- Insufficient Balance Test ---

Transaction Failed: Insufficient balance for this payment.

Transaction processing completed.

----- Wallet Details -----

User Name : Lalitha

Mobile Number: 9876732100

UPI ID : lalitha@upi

Balance : ₹3500.0